Algebra II Quadratics and Optimization Cheatsheet

Module 3

Forms

Standard:

$$f(x) = ax^2 + bx + c$$

Vertex:

$$f(x) = a(x-h)^2 + k$$

Factored:

$$f(x) = a(x-r_1)(x-r_2)$$

Vertex

From standard form:

$$x = -\frac{b}{2a}$$

$$y = f\left(-\frac{b}{2a}\right)$$

From vertex form:

$$(h, k)$$

Axis of symmetry:

$$x = h$$

Solving

Factoring:

$$AB = 0 \Rightarrow A = 0 \text{ or } B = 0$$

Completing the square:

$$x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2$$

Quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant

$$D = b^2 - 4ac$$

$$\begin{array}{ll} D > 0 & \text{two distinct real roots} \\ D = 0 & \text{one repeated real root} \\ D < 0 & \text{two complex conjugate roots} \end{array}$$

Complex Solutions

$$i^2 = -1, \qquad \sqrt{-a} = i\sqrt{a}$$

If coefficients are real and $a + bi$ is a root, then $a - bi$ is also a root.

Optimization

For a quadratic model:

$$f(x) = ax^2 + bx + c$$

The maximum or minimum occurs at:

$$x = -\frac{b}{2a}$$

$$a > 0 \Rightarrow \text{minimum}, \qquad a < 0 \Rightarrow \text{maximum}$$

Revenue form:

$$R(x) = x \cdot p(x)$$

Quick Checks

- Choose the form that reveals the feature needed.
- Vertex form reveals vertex and transformations.
- Factored form reveals zeros.
- Standard form reveals y-intercept and supports the quadratic formula.